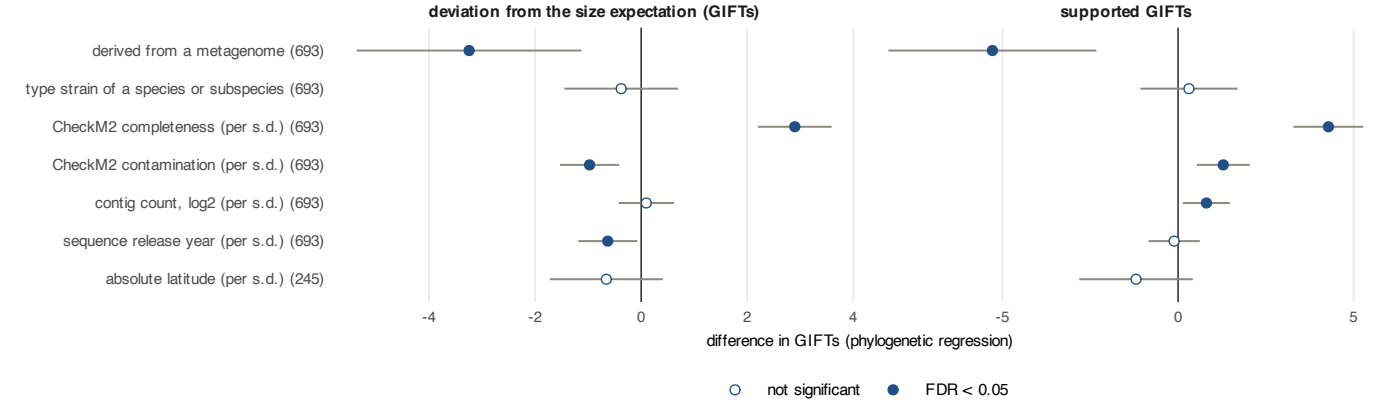


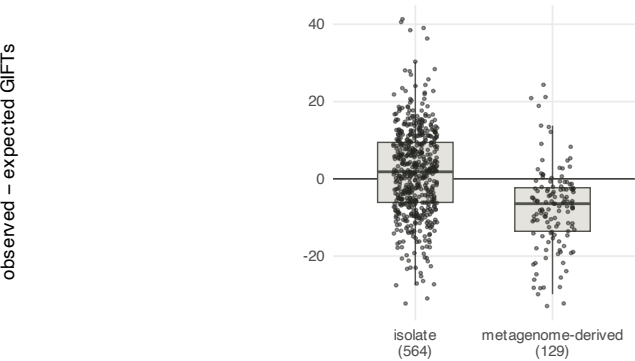
Encoded repertoire by assembly annotation

a: annotations fitted jointly in a phylogenetic regression (genomes in brackets; latitude fitted alone on the genomes with coordinates), points with 95% intervals. b to d: deviation from the Figure S9 size expectation.

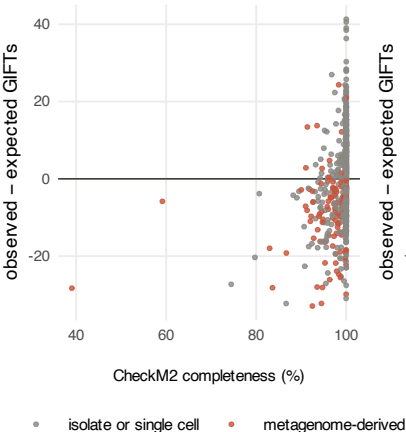
a Assembly annotations



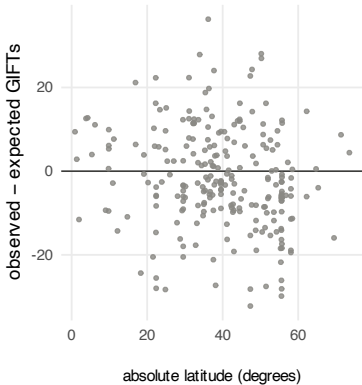
b Genome category



c Estimated completeness



d Latitude



Completeness and contamination are CheckM2 estimates, which are themselves marker-based. A difference describes encoded, curated capability in this panel, not activity or phenotype.